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Inventor(s)	Goodman; Brandon C.

Cementing stage tool and associated methods

Abstract

A method of cementing a tubular string in a wellbore can include applying a predetermined pressure differential from a flow passage extending axially through the tubular string to an annulus surrounding the tubular string, thereby opening a rupture disk of a cementing stage tool connected in the tubular string, and then flowing a fluid through the flow passage and into the annulus via the rupture disk, thereby displacing an opening plug into engagement with an opening sleeve of the cementing stage tool. A cementing stage tool can include a longitudinal flow passage, an outer housing assembly, an opening sleeve that prevents fluid flow between the flow passage and a housing port in a run-in configuration, and a rupture disk that permits fluid flow between the flow passage and the housing port in response to a predetermined pressure differential applied from the flow passage to the housing port.

Inventors:	Goodman; Brandon C. (Houston, TX)
Applicant:	WEATHERFORD TECHNOLOGY HOLDINGS, LLC (Houston, TX)
Family ID:	96264532
Assignee:	WEATHERFORD TECHNOLOGY HOLDINGS, LLC (Houston, TX)
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Primary Examiner: Fuller; Robert E

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of the filing date of U.S. provisional application No. 63/619,507 filed on 10 Jan. 2024. The entire disclosure of the prior application is incorporated herein by this reference in its entirety for all purposes.

BACKGROUND

(1) This disclosure relates generally to equipment utilized and operations performed in conjunction with a subterranean well and, in an example described below, more particularly provides a cementing stage tool and associated methods.

(2) A cementing stage tool can be used to control communication between a tubular string bore and an annulus surrounding the tubular string in a cementing operation. It is very important for such communication between the tubular string bore and the annulus to be selectively permitted or prevented at various different points in the cementing operation.

(3) It will, therefore, be readily appreciated that improvements are continually needed in the art of designing, constructing and utilizing cementing stage tools. The present disclosure provides such improvements to the art, which improvements may be useful in a wide variety of different types of well cementing operations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a representative partially cross-sectional view of an example of a well system and associated method which can embody principles of this disclosure.

(2) FIG. 2 is a representative cross-sectional view of an example of a cementing stage tool that may be used in the FIG. 1 system and method, the cementing stage tool being depicted in a run-in configuration.

(3) FIG. 3 is a representative cross-sectional view of the cementing stage tool in a circulate configuration.

(4) FIG. 4 is a representative cross-sectional view of the cementing stage tool in a cementing configuration.

(5) FIG. 5 is a representative cross-sectional view of the cementing stage tool in an end-of-stage configuration.

(6) FIG. 6 is a representative cross-sectional view of the cementing stage tool in a closing sleeve unlocked configuration.

(7) FIG. 7 is a representative cross-sectional view of the cementing stage tool in a closed configuration.

DETAILED DESCRIPTION

(8) Representatively illustrated in FIG. 1 is a system **10** for use with a subterranean well, and an associated method, which can embody principles of this disclosure. However, it should be clearly understood that the system **10** and method are merely one example of an application of the principles of this disclosure in practice, and a wide variety of other examples are possible. Therefore, the scope of this disclosure is not limited at all to the details of the system **10** and method described herein and/or depicted in the drawings.

(9) In the FIG. 1 example, it is desired to place cement into an annulus **12** surrounding a tubular

string **14**. As depicted in FIG. **1**, the annulus **12** is formed radially between the tubular string **14** and a wellbore **16** drilled into the earth. In other examples, the annulus **12** may be formed between the tubular string **14** and another tubular string (not shown) outwardly surrounding the tubular string **14**.

(10) In various examples, the tubular string **14** may be of the types known to those skilled in the art as casing, liner, tubing or pipe. The scope of this disclosure is not limited to use of any particular type of tubular string.

(11) An inner flow passage **18** extends longitudinally through the tubular string **14**. A cementing stage tool **20** is connected in the tubular string **14** to control fluid communication between the flow passage **18** and the annulus **12** during the cementing operation.

(12) To actuate the cementing stage tool **20**, so that fluid communication is permitted between the flow passage **18** and the annulus **12**, it is desired to displace an opening plug **22** with circulating fluid flow **24** through the flow passage to the cementing stage tool. However, in the FIG. **1** example, such circulating fluid flow **24** is initially prevented by either a bridge plug **26** set in the flow passage **18** downhole of the cementing stage tool **20**, or a packer **28** set in the annulus **12** downhole of the cementing stage tool.

(13) Fortunately, the cementing stage tool **20** includes features that enable fluid communication to be established between the flow passage **18** and the annulus **12**, so that the plug **22** can be displaced to the cementing stage tool with the fluid flow **24**, even though circulating fluid flow is not initially permitted from the flow passage to the annulus through a sidewall of the cementing stage tool when it is initially deployed into the wellbore **16** with the tubular string **14**.

(14) Referring additionally now to FIG. **2**, a cross-sectional view of an example of the cementing stage tool **20** is representatively illustrated. The FIG. **2** cementing stage tool **20** may be used with the FIG. **1** system **10** and method, or it may be used with other systems and methods. For convenience, the cementing stage tool **20** is described below as it may be used with the FIG. **1** system **10** and method.

(15) As depicted in FIG. **2**, the cementing stage tool **20** is in a run-in configuration. The cementing stage tool **20** is in this configuration when it is initially connected in the tubular string **14** and deployed into the wellbore **16** in the FIG. **1** system **10** and method.

(16) In the FIG. **2** example, the cementing stage tool **20** includes a generally tubular outer housing assembly **30**, an opening sleeve **32**, a closing sleeve **34** and a release sleeve **36**. As described more fully below, the opening and closing sleeves **32**, **34** are used to selectively permit and prevent fluid communication between the flow passage **18** and the annulus **12** via one or more housing ports **38** formed through sidewall of the outer housing assembly **30**.

(17) The outer housing assembly **30** in this example includes upper and lower connectors **41**, **43** for connecting the cementing stage tool **20** in the tubular string **14**. For this purpose, the upper and lower connectors **41**, **43** can include threads, seals, etc. to secure and seal the cementing stage tool **20** in the tubular string **14**, while enabling the flow passage **18** to extend axially through the tool and isolating the flow passage from the annulus **12** at the tool.

(18) The opening sleeve **32** is initially in a closed position in which it blocks fluid flow between the flow passage **18** and the annulus **12** via the housing ports **38**. The opening sleeve **32** is initially retained in the closed position by shear members **40** (such as, shear screws, shear pins, a shear ring, etc.) or other releasable retainer members (such as, a snap ring, collets, etc.). In the FIG. **2** example, the shear members **40** extend through a sidewall of the opening sleeve **32** and into an abutment sleeve **42** threaded into a lower end of the closing sleeve **34**.

(19) An annulus **44** is isolated between upper and lower seals **46**, **48**. The seal **46** seals between the opening sleeve **32** and the closing sleeve **34**, and the seal **48** seals between the opening sleeve and the abutment sleeve **42**, respectively above and below the annulus **44**. Similarly, upper and lower seals **50**, **52** seal between the closing sleeve **34** and the outer housing assembly **30**, respectively above and below the housing ports **38**.

(20) Circulation ports **54** are formed through the sidewall of the opening sleeve **32** axially between the seals **46**, **48**. Rupture disks **56** initially prevent fluid flow through the circulation ports **54**. However, the rupture disks **56** are selected, so that they will open to permit fluid communication between the flow passage **18** and the annulus **12** when a predetermined pressure differential is applied from the flow passage to the annulus.

(21) The closing sleeve **34** includes closing sleeve ports **58** formed through a sidewall of the closing sleeve. In the run-in configuration of the cementing stage tool **20**, the closing sleeve ports **58** are in fluid communication with the housing ports **38** and the annulus **44**. The closing sleeve **34** is releasably retained in this position by a lock mechanism **60**.

(22) The lock mechanism **60** includes radially retractable lock rings **62** radially outwardly supported in engagement with an annular recess **64** formed in the outer housing assembly **30**. The lock rings **62** are outwardly supported by the release sleeve **36**, which is releasably retained in this position by shear members **66** (such as, shear screws, shear pins, a shear ring, etc.) or other releasable retainer members (such as, a snap ring, collets, etc.). The shear members **66** extend through a sidewall of the release sleeve **36** and into the sidewall of the closing sleeve **34**.

(23) Referring additionally now to FIG. 3, a cross-sectional view of the cementing stage tool **20** is representatively illustrated. As depicted in FIG. 3, the cementing stage tool **20** is in a circulate configuration in which fluid flow **24** is permitted from the flow passage **18** to the annulus **12**.

(24) The predetermined pressure differential has been applied from the flow passage **18** to the annulus **12** to thereby open the rupture disks **56**. Fluid communication is, thus, permitted between the flow passage **18** and the annulus **12** via the circulation ports **54**, the annulus **44**, the closing sleeve ports **58** and the housing ports **38**.

(25) The circulating fluid flow **24** can be used to displace the plug **22** through the tubular string **14** to the cementing stage tool **20**. Note that the fluid flow **24** passes through the relatively small flow area rupture disks **56**, and so a flow rate of the fluid flow **24** will be relatively restricted. This restriction of fluid flow can be used to provide an indication of the cementing stage tool's circulate configuration (e.g., due to the circulation fluid flow **24** being permitted, but at a relatively low flow rate). Thus, an operator at the surface will be able to know positively that the cementing stage tool **20** is in the circulate configuration, instead of some other configuration.

(26) Note that the rupture disks **56** are one example of a pressure operated flow control device that may be used to selectively permit fluid communication through the sidewall of the opening sleeve **32** in response to the predetermined pressure differential applied from the flow passage **18** to the annulus **12**. In other examples, other types of flow control devices (such as, a sliding sleeve valve, a relief valve, etc.) may be used instead of the rupture disks **56**.

(27) Referring additionally now to FIG. 4, a cross-sectional view of the cementing stage tool **20** is representatively illustrated. As depicted in FIG. 4, the cementing stage tool **20** is in a cementing configuration.

(28) In the FIG. 4 example, the plug **22** has been displaced into the cementing stage tool **20** with the circulating fluid flow **24** (see FIG. 3). The plug **22** has been displaced into sealing contact with an annular seat **68** formed in an upper end of the opening sleeve **32**. The plug **22** now blocks fluid communication through the flow passage **18** at the opening sleeve **32**.

(29) As depicted in FIG. 4, the plug **22** is in the form of a ball or sphere. In other examples, the plug **22** could comprise a dart, a cylinder, or another shape or configuration suitable for blocking fluid flow through the opening sleeve **32**.

(30) After the plug **22** engages the seat **68**, fluid pressure in the flow passage **18** uphole of the plug is increased until a predetermined pressure differential is applied across the plug and seat. The pressure differential is from the flow passage **18** uphole of the plug **22** to the flow passage downhole of the plug (including to the circulation ports **54**, the annulus **44** (see FIG. 3), the closing sleeve ports **58**, the housing ports **38** and the annulus **12**). Note that this pressure differential may be greater than, less than or equal to the pressure differential described above for opening the

rupture disks **56**.

(31) The predetermined pressure differential applied across the plug **22** and seat **68** causes the shear members **40** to shear, so that the opening sleeve **32** displaces downward in the housing assembly **30** to an open position as depicted in FIG. **4**. The downward displacement of the opening sleeve **32** is limited by contact with an upper end of the abutment sleeve **42**.

(32) With the opening sleeve **32** in its FIG. **4** open position, fluid communication is permitted between the flow passage **18** and the annulus **12** via the closing sleeve ports **58** and the housing ports **38**. However, fluid flow between the flow passage **18** and the annulus **12** will no longer pass through the sidewall of the opening sleeve **32** via the circulation ports **54** and the rupture disks **56**.

(33) With the opening sleeve **32** in its open position, an increased rate of fluid flow will be permitted from the flow passage **18** to the annulus **12**. This is due to the fluid flow being able to pass from the flow passage **18** to the annulus **12** via the closing sleeve ports **58** and the housing ports **38**, without having to pass through the relatively restrictive rupture disks **56**. The operator at the surface will observe the increased flow rate as a positive indication that the opening sleeve **32** has shifted to its open position.

(34) Cement **70** can now be flowed from the flow passage **18** to the annulus **12** via the closing sleeve ports **58** and the housing ports **38**. Note that, due to the plug **22** blocking flow through the flow passage **18** at the opening sleeve **32**, the cement **70** does not flow into the flow passage **18** downhole of the plug and the opening sleeve. This advantageously reduces the amount of cement **70** that will later need to be drilled or milled out of the flow passage **18**.

(35) As a contingency measure, if it previously had not been possible to open the rupture disks **56** in order to circulate the plug **22** to the opening sleeve **32**, it would still be possible to shift the opening sleeve to its open position by applying a predetermined pressure differential from the flow passage **18** to the annulus **12**. This is due to a differential piston area on the opening sleeve **32** resulting from the difference in diameter between the upper and lower seals **46**, **48**. This differential piston area enables the predetermined pressure differential from the flow passage **18** to the annulus **12** to apply a downwardly directed biasing force to the opening sleeve **32**, and thereby shear the shear members **40** and displace the opening sleeve downward to the open position.

(36) Referring additionally now to FIG. **5**, a cross-sectional view of the cementing stage tool **20** is representatively illustrated. As depicted in FIG. **5**, the cementing stage tool **20** is in an end-of-stage configuration.

(37) In the FIG. **5** example, the cement **70** has been placed in the annulus **12**. There is also a relatively small amount of the cement **70** in the flow passage **18** uphole of the plug **22** and seat **68**. However, there is no cement in the flow passage **18** downhole of the plug **22** and seat **68**.

(38) A closing plug **72** has been displaced through the flow passage **18** following the cement **70**. The plug **72** is sealingly engaged in a bore **74** of the upper connector **41**. As depicted in FIG. **5**, the plug **72** is in the form of a dart, but other types of plugs may be used in other examples. In other examples, the plug **72** could be sealingly engaged with another component of the cementing stage tool **20** (such as, the closing sleeve **34** or the release sleeve **36**).

(39) Referring additionally now to FIG. **6**, a cross-sectional view of the cementing stage tool **20** is representatively illustrated. As depicted in FIG. **6**, the cementing stage tool **20** is in a closing sleeve **34** unlocked configuration.

(40) In the unlocked configuration, a predetermined pressure differential has been applied across the plug **72** (from a portion of the flow passage **18** uphole of the plug **72** to a portion of the flow passage downhole of the plug) to thereby apply a downwardly directed biasing force to the release sleeve **36**. This biasing force shears the shear members **66** and displaces the release sleeve **36** downward.

(41) In its downwardly displaced unlocked position, the release sleeve **36** no longer outwardly supports the lock rings **62** in engagement with the annular recess **64**. As a result, the lock rings **62** can retract radially inward. With the lock rings **62** retracted radially inward, the closing sleeve **34** is

no longer prevented from displacing axially relative to the outer housing assembly **30**.

(42) Referring additionally now to FIG. 7, a cross-sectional view of the cementing stage tool **20** is representatively illustrated. As depicted in FIG. 7, the cementing stage tool **20** is in a closed configuration.

(43) Once the release sleeve **36** is displaced to its unlocked position as described above, the downwardly directed biasing force due to the pressure differential across the plug **72** will cause the closing sleeve **34** to displace downward to its closed position as depicted in FIG. 7. In this closed position, seals **50**, **76** straddling the housing ports **38** prevent fluid communication between the flow passage **18** and the annulus **12** via the housing ports. The seals **50**, **76** seal between the closing sleeve **34** and the outer housing assembly **30** on opposite sides of the housing ports **38**.

(44) If it is desired to later establish fluid communication longitudinally through the cementing stage tool **20**, the flow passage **18** can be drilled or milled out. Preferably, all of the internal components of the cementing stage tool **20** (including but not limited to the plug **72**, the plug **22**, the opening sleeve **32** and the rupture disks **56**) are made of materials that can be readily drilled or milled through.

(45) It may now be fully appreciated that the above disclosure provides to the art an improved cementing stage tool **20** and associated methods. In examples described above, the cementing stage tool **20** enables circulating fluid flow **24** to be established at the tool, even though a bridge plug **26** has been set in the flow passage **18** downhole of the tool, or a packer **28** has been set in the annulus **12** downhole of the tool. In this way, the plug **22** can be displaced to the cementing stage tool **20** prior to flowing cement **70** to the tool, thereby preventing the cement from flowing into the flow passage **18** downhole of the tool.

(46) The above disclosure provides to the art a method of cementing a tubular string **14** in a wellbore **16**. In one example, the method can comprise: applying a first predetermined pressure differential from a flow passage **18** extending axially through the tubular string **14** to an annulus **12** surrounding the tubular string **14**, thereby opening a rupture disk **56** of a cementing stage tool **20** connected in the tubular string **14**; and then flowing a fluid through the flow passage **18** and into the annulus **12** via the rupture disk **56**, thereby displacing an opening plug **22** into engagement with an opening sleeve **32** of the cementing stage tool **20**.

(47) The method may include, after the flowing step, applying a second predetermined pressure differential from the flow passage **18** to the annulus **12**, thereby displacing the opening sleeve **32** to an open position. A rate of fluid flow from the flow passage **18** to the annulus **12** after the rupture disk **56** opening and prior to the step of displacing the opening sleeve **32** to the open position may be less than a rate of fluid flow from the flow passage **18** to the annulus **12** after the step of displacing the opening sleeve **32** to the open position.

(48) The method may include, after the step of displacing the opening sleeve **32** to the open position, flowing cement **70** from an uphole portion of the flow passage **18** to the annulus **12**. The opening plug **22** prevents the cement **70** from flowing into a downhole portion of the flow passage **18**, the downhole portion of the flow passage **18** being disposed opposite the opening plug **22** from the uphole portion of the flow passage **18**.

(49) During the first pressure differential applying step, the flow passage **18** may be isolated from the annulus **12** downhole of the cementing stage tool **20**.

(50) The step of opening the rupture disk **56** may include permitting fluid communication through at least one circulation port **54** formed through a sidewall of the opening sleeve **32**. The step of opening the rupture disk **56** may also include permitting fluid communication between the flow passage **18** and the annulus **12** via the circulation port **54**, a housing port **38** formed through a sidewall of an outer housing assembly **30**, and a closing sleeve port **58** formed through a sidewall of a closing sleeve **34** positioned between the opening sleeve **32** and the outer housing assembly **30**.

(51) The above disclosure also provides to the art a cementing stage tool **20** for use with a

subterranean well. In one example, the cementing stage tool **20** can comprise: a flow passage **18** extending longitudinally through the cementing stage tool **20**; an outer housing assembly **30** including a housing port **38** extending through a sidewall of the outer housing assembly **30**; an opening sleeve **32** releasably secured against axial displacement in the outer housing assembly **30**, the opening sleeve **32** being configured to prevent fluid flow between the flow passage **18** and the housing port **38** in a run-in configuration of the cementing stage tool **20**; and a rupture disk **56** configured to prevent fluid flow between the flow passage **18** and the housing port **38**. The rupture disk **56** is configured to permit fluid flow between the flow passage **18** and the housing port **38** in response to a predetermined pressure differential applied from the flow passage **18** to the housing port **38**.

(52) The rupture disk **56** may be disposed in a sidewall of the opening sleeve **32**. The rupture disk **56** may prevent fluid flow through a circulation port **54** formed through a sidewall of the opening sleeve **32**, the circulation port **54** being positioned between first and second seals **46**, **48** that isolate an annulus **44** between the opening sleeve **32** and a closing sleeve **34** in the outer housing assembly **30**.

(53) The first seal **46** may have a diameter which is greater than a diameter of the second seal **48**. In some examples, the first seal **46** may have a diameter which is less than or equal to a diameter of the second seal **48**. The closing sleeve **34** may have a closing sleeve port **58** that permits fluid communication between the annulus **44** and the housing port **38**.

(54) The opening sleeve **32** may include a seat **68** configured for engagement with an opening plug **22**. The opening sleeve **32** may be configured to displace to an open position in which fluid flow between the flow passage **18** and the housing port **38** is permitted in response to a predetermined pressure differential applied across the opening plug **22**. The opening plug **22** is sealed with the opening sleeve **32** in an example described above.

(55) The rupture disk **56** may be configured to permit the fluid flow **24** between the flow passage **18** and the housing port **38** at a reduced flow rate as compared to the fluid flow **24** between the flow passage **18** and the housing port **38** permitted with the opening sleeve **32** in the open position.

(56) Another method described above for cementing a tubular string **14** in a wellbore **16** can comprise: applying a first predetermined pressure differential from a flow passage **18** extending axially through the tubular string **14** to an annulus **12** surrounding the tubular string **14**, thereby permitting fluid communication between the flow passage **18** and the annulus **12** via a circulation port **54** formed through a sidewall of an opening sleeve **32** of a cementing stage tool **20**; and then applying a second predetermined pressure differential from the flow passage **18** to the annulus **12**, thereby displacing the opening sleeve **32** to an open position and permitting fluid communication between the flow passage **18** and the annulus **12** without flowing through the circulation port **54**.

(57) The method may include, after the step of applying the first predetermined pressure, then flowing a fluid through the flow passage **18** and into the annulus **12** via the circulation port **54**, thereby displacing an opening plug **22** into engagement with the opening sleeve **32**.

(58) The step of permitting fluid communication between the flow passage **18** and the annulus **12** via the circulation port **54** may comprise opening a rupture disk **56**.

(59) A rate of fluid flow **24** from the flow passage **18** to the annulus **12** after step of the permitting fluid communication between the flow passage **18** and the annulus **12** via the circulation port **54** and prior to the step of displacing the opening sleeve **32** to the open position may be less than a rate of fluid flow from the flow passage **18** to the annulus **12** after the step of displacing the opening sleeve **32** to the open position.

(60) During the first pressure differential applying step, the flow passage **18** may be isolated from the annulus **12** downhole of the cementing stage tool **20**.

(61) The step of permitting fluid communication between the flow passage **18** and the annulus **12** via the circulation port **54** may include permitting fluid communication through a housing port **38** formed through an outer housing assembly **30**, and a closing sleeve port **58** formed through a

sidewall of a closing sleeve 34 positioned between the opening sleeve 32 and the outer housing assembly 30.

(62) Although various examples have been described above, with each example having certain features, it should be understood that it is not necessary for a particular feature of one example to be used exclusively with that example. Instead, any of the features described above and/or depicted in the drawings can be combined with any of the examples, in addition to or in substitution for any of the other features of those examples. One example's features are not mutually exclusive to another example's features. Instead, the scope of this disclosure encompasses any combination of any of the features.

(63) Although each example described above includes a certain combination of features, it should be understood that it is not necessary for all features of an example to be used. Instead, any of the features described above can be used, without any other particular feature or features also being used.

(64) It should be understood that the various embodiments described herein may be utilized in various orientations, such as inclined, inverted, horizontal, vertical, etc., and in various configurations, without departing from the principles of this disclosure. The embodiments are described merely as examples of useful applications of the principles of the disclosure, which is not limited to any specific details of these embodiments.

(65) In the above description of the representative examples, directional terms (such as “above,” “below,” “upper,” “lower,” “upward,” “downward,” etc.) are used for convenience in referring to the accompanying drawings. However, it should be clearly understood that the scope of this disclosure is not limited to any particular directions described herein.

(66) The terms “including,” “includes,” “comprising,” “comprises,” and similar terms are used in a non-limiting sense in this specification. For example, if a system, method, apparatus, device, etc., is described as “including” a certain feature or element, the system, method, apparatus, device, etc., can include that feature or element, and can also include other features or elements. Similarly, the term “comprises” is considered to mean “comprises, but is not limited to.”

(67) Of course, a person skilled in the art would, upon a careful consideration of the above description of representative embodiments of the disclosure, readily appreciate that many modifications, additions, substitutions, deletions, and other changes may be made to the specific embodiments, and such changes are contemplated by the principles of this disclosure. For example, structures disclosed as being separately formed can, in other examples, be integrally formed and vice versa. Accordingly, the foregoing detailed description is to be clearly understood as being given by way of illustration and example only, the spirit and scope of the invention being limited solely by the appended claims and their equivalents.

Claims

1. A method of cementing a tubular string in a wellbore, the method comprising: applying a first predetermined pressure differential between a flow passage extending axially through the tubular string and an annulus surrounding the tubular string, thereby permitting fluid communication between the flow passage and the annulus via a circulation port formed through a sidewall of an opening sleeve of a cementing stage tool; and then applying a second predetermined pressure differential between the flow passage and the annulus, thereby displacing the opening sleeve to an open position and permitting fluid communication between the flow passage and the annulus without flowing through the circulation port.
2. The method of claim 1, further comprising, after the applying the first predetermined pressure, then flowing a fluid through the flow passage and into the annulus via the circulation port, thereby displacing an opening plug into engagement with the opening sleeve.
3. The method of claim 1, in which the permitting fluid communication between the flow passage

and the annulus via the circulation port comprises opening a rupture disk.

4. The method of claim 1, in which a rate of fluid flow from the flow passage to the annulus after the permitting fluid communication between the flow passage and the annulus via the circulation port and prior to the displacing the opening sleeve to the open position is less than a rate of fluid flow from the flow passage to the annulus after the displacing the opening sleeve to the open position.

5. The method of claim 1, in which, during the first pressure differential applying, the flow passage is isolated from the annulus downhole of the cementing stage tool.

6. The method of claim 1, in which the permitting fluid communication between the flow passage and the annulus via the circulation port comprises permitting fluid communication through a housing port formed through an outer housing assembly, and a closing sleeve port formed through a sidewall of a closing sleeve positioned between the opening sleeve and the outer housing assembly.
